

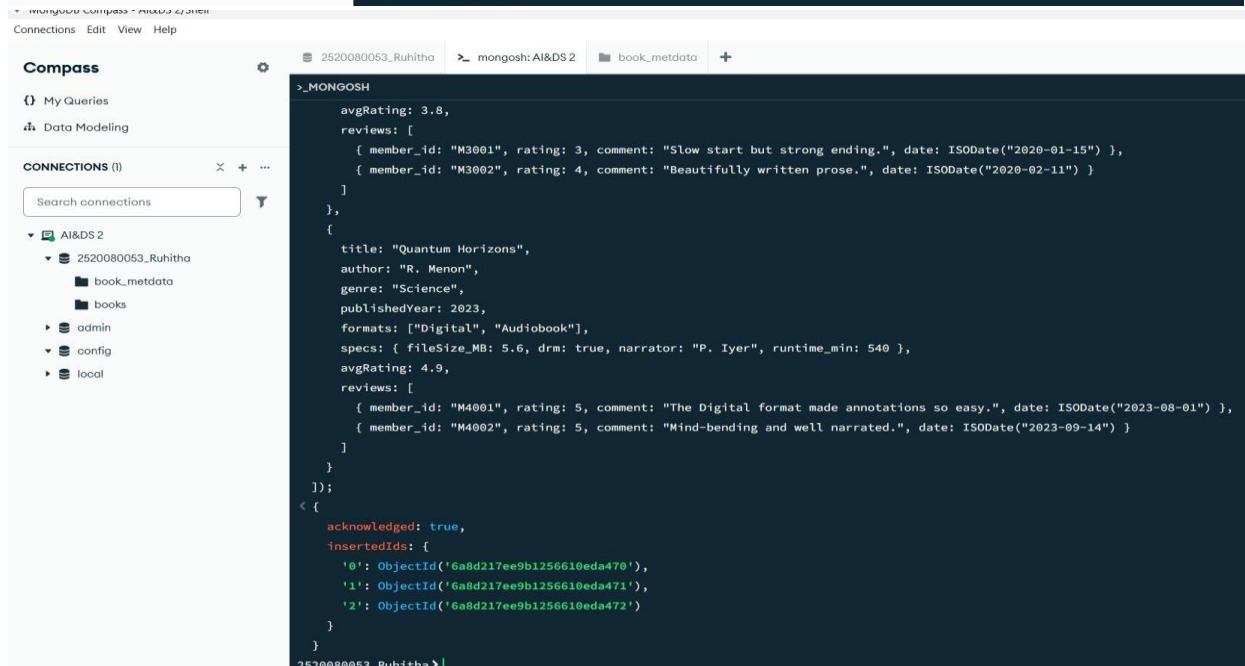
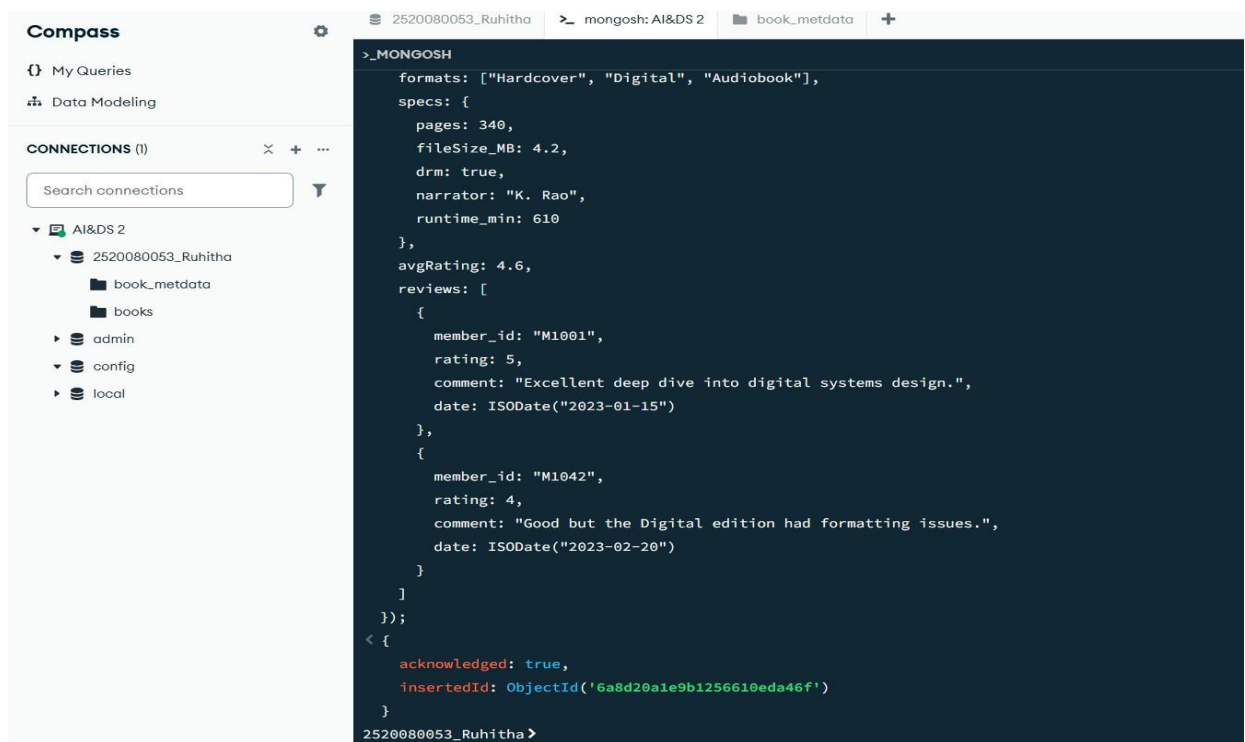
PRACTICAL

WEEK 3

2520090046

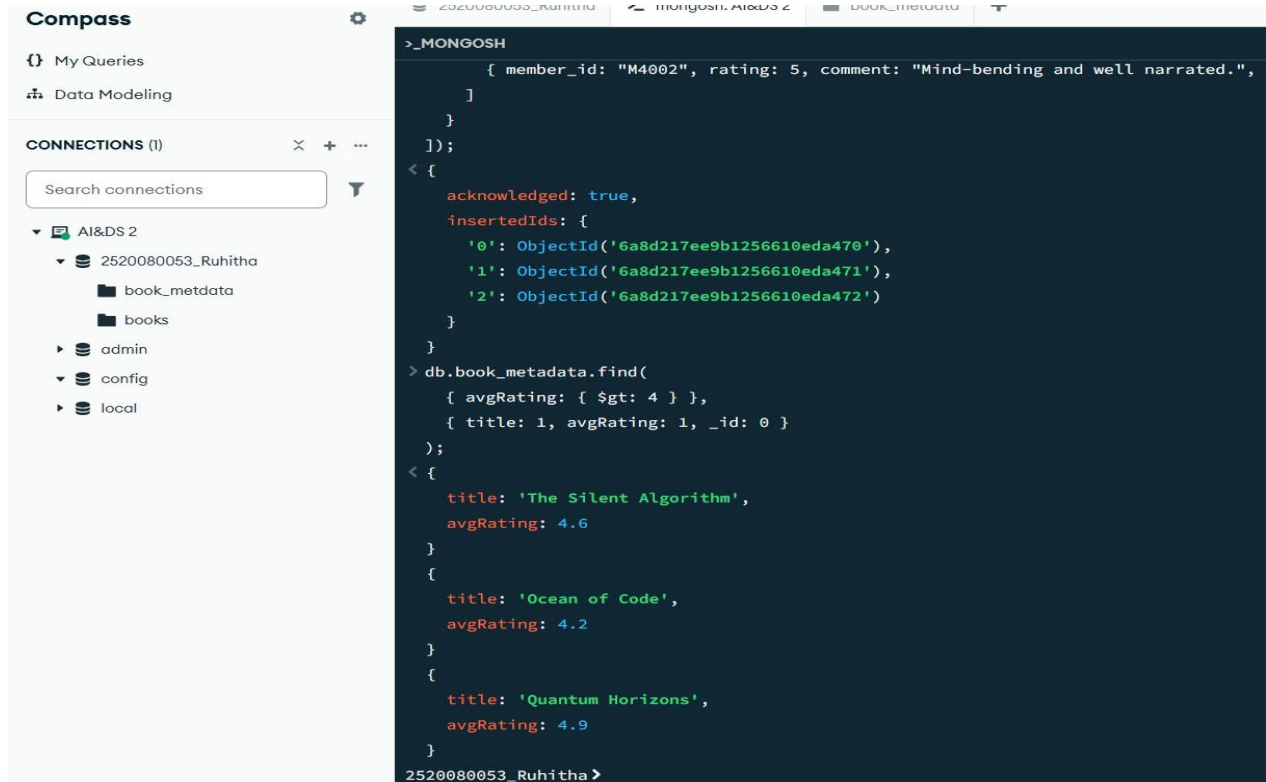
G SHIVANI

Task 1: Document Modelling



Task 2: Advanced Querying

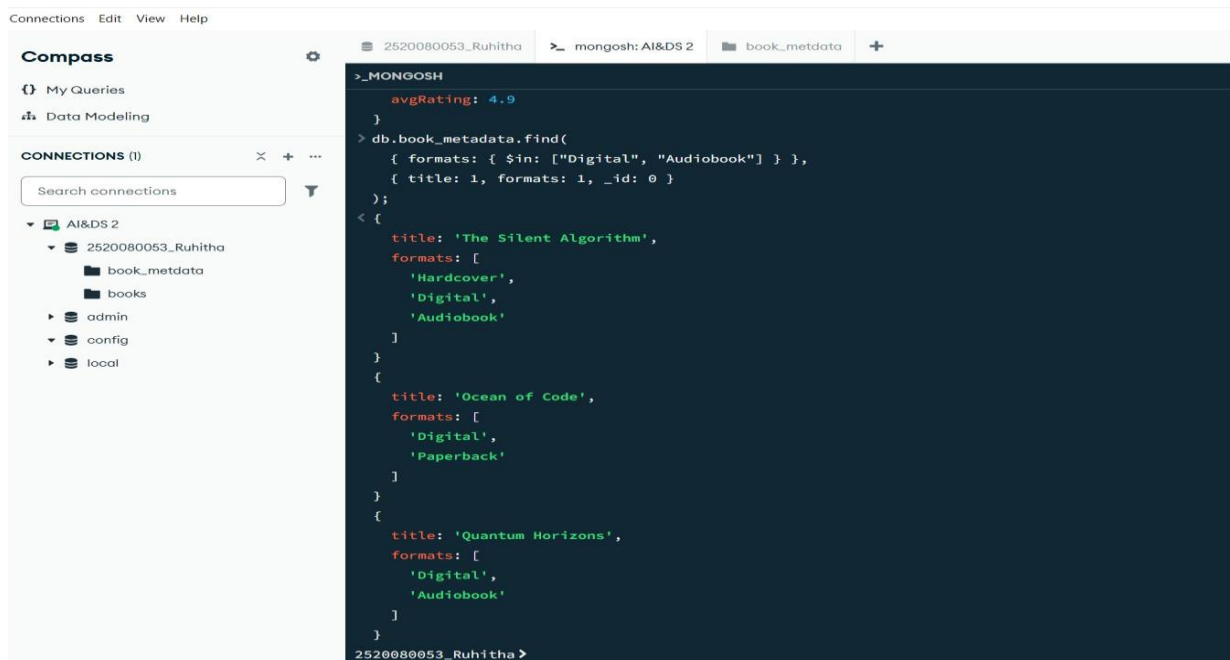
a) Books rated above 4 (\$gt)



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the database 'AI&DS 2' and its collections, including 'book_metadata'. The main editor shows a MongoDB query in the 'MONGOSH' shell:

```
>_MONGOSH
{ member_id: "M4002", rating: 5, comment: "Mind-bending and well narrated.",
}
});
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8d217ee9b1256610eda470'),
    '1': ObjectId('6a8d217ee9b1256610eda471'),
    '2': ObjectId('6a8d217ee9b1256610eda472')
  }
}
> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
);
< {
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  avgRating: 4.2
}
{
  title: 'Quantum Horizons',
  avgRating: 4.9
}
}
```

b) Books available in specific formats (\$in)



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the database 'AI&DS 2' and its collections, including 'book_metadata'. The main editor shows a MongoDB query in the 'MONGOSH' shell:

```
>_MONGOSH
avgRating: 4.9
}
> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
);
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
{
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
}
```

c) Combine both with \$and (implicit)

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

- 2520080053_Ruhitha
 - book_metdata
 - books
- admin
- config
- local

2520080053_Ruhitha mongosh: AI&DS 2 book_metdata +

```
>_MONGOSH
}
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
});
< {
  _id: ObjectId('6a8d20a1e9b1256610eda46f'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.'
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

- 2520080053_Ruhitha
 - book_metdata
 - books
- admin
- config
- local

2520080053_Ruhitha mongosh: AI&DS 2 book_metdata +

```
>_MONGOSH
genre: 'Science',
publishedYear: 2023,
formats: [
  'Digital',
  'Audiobook'
],
specs: {
  fileSize_MB: 5.6,
  drm: true,
  narrator: 'P. Iyer',
  runtime_min: 540
},
avgRating: 4.9,
reviews: [
  {
    member_id: 'M4001',
    rating: 5,
    comment: 'The Digital format made annotations so easy.',
    date: 2023-08-01T00:00:00.000Z
  },
  {
    member_id: 'M4002',
    rating: 5,
    comment: 'Mind-bending and well narrated.',
    date: 2023-09-14T00:00:00.000Z
  }
]
}
```

d) Regex query — books with "Digital" format

Connections Edit View Help

2520080053_Ruhitha mongosh: AI&DS 2 book_metadata +

Compass

{ } My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

- 2520080053_Ruhitha
 - book_metadata
 - books
- admin
- config
- local

```
>_MONGOSH
> db.book_metadata.find(
  { formats: { $regex: "Digital", $options: "i" } },
  { title: 1, formats: 1 }
);
< {
  _id: ObjectId('6a8d20a1e9b1256610eda46f'),
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  _id: ObjectId('6a8d217ee9b1256610eda470'),
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
{
  _id: ObjectId('6a8d217ee9b1256610eda472'),
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
```

e) Regex query — reviews mentioning "Digital" in comments (nested field)

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

- 2520080053_Ruhitha
 - book_metadata
 - books
- admin
- config
- local

```
>_MONGOSH
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } },
  { title: 1, "reviews.$": 1 }
);
< {
  _id: ObjectId('6a8d20a1e9b1256610eda46f'),
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a8d217ee9b1256610eda470'),
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    }
  ]
}
}
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

- 2520080053_Ruhitha
 - book_metadata
 - books
- admin
- config
- local

```
>_MONGOSH
    date: 2023-01-15T00:00:00.000Z
  }
]
}
{
  _id: ObjectId('6a8d217ee9b1256610eda470'),
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    }
  ]
}
}
{
  _id: ObjectId('6a8d217ee9b1256610eda472'),
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: 2023-08-01T00:00:00.000Z
    }
  ]
}
}
}
2520080053_Ruhitha>
```

Task 3: Aggregation Pipeline — "Top Rated" Report

Connections Edit View Help

2520080053_Ruhitha mongosh: AI&DS 2 book_metdata +

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

- 2520080053_Ruhitha
 - book_metdata
 - books
- admin
- config
- local

```
>_MONGOSH
> db.book_metdata.aggregate([

  { $match: { publishedYear: { $gt: 2020 } } },

  { $unwind: "$reviews" },

  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },

  { $sort: { avgReviewRating: -1 } },

  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: { $round: ["$avgReviewRating", 2] },
      totalReviews: 1
    }
  }
])
```

2520080053_Ruhitha mongosh: AI&DS 2 book_metdata +

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

- 2520080053_Ruhitha
 - book_metdata
 - books
- admin
- config
- local

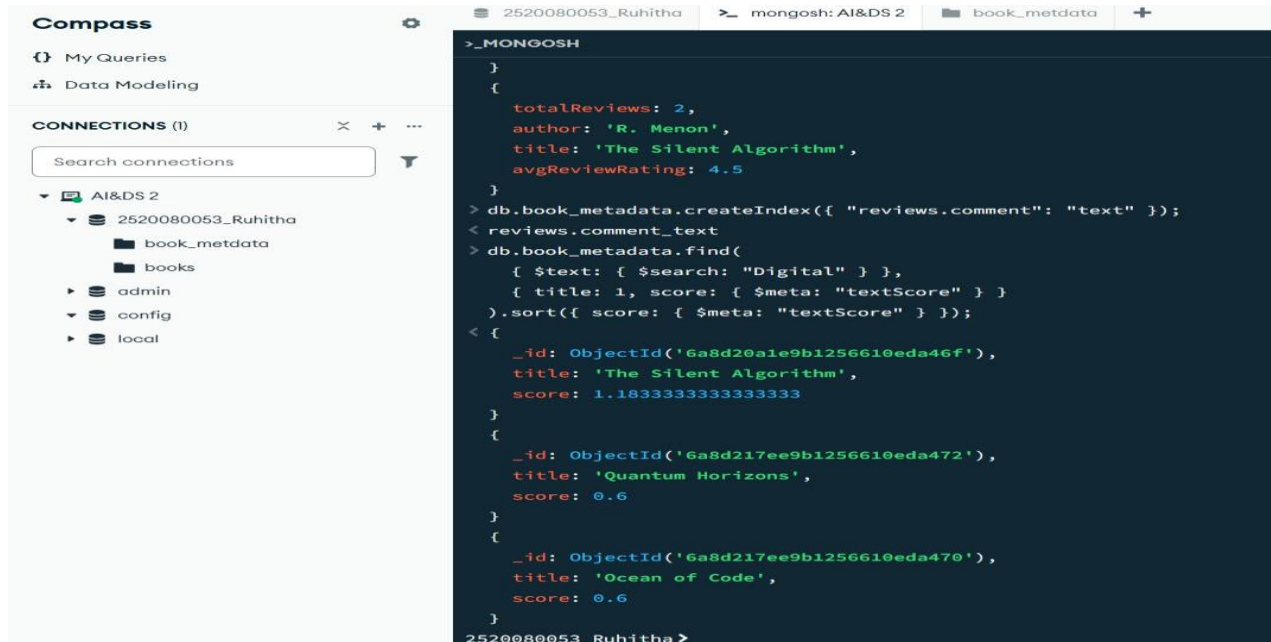
```
>_MONGOSH
{
  $project: {
    _id: 0,
    title: "$_id",
    author: 1,
    avgReviewRating: { $round: ["$avgReviewRating", 2] },
    totalReviews: 1
  }
}
]);
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
}
2520080053_Ruhitha >
```

Task 4: Indexing for Speed

a) Create a text index on review comments

```
}  
> db.book_metadata.createIndex({ "reviews.comment": "text" });  
< reviews.comment_text  
2520080053_Ruhitha >
```

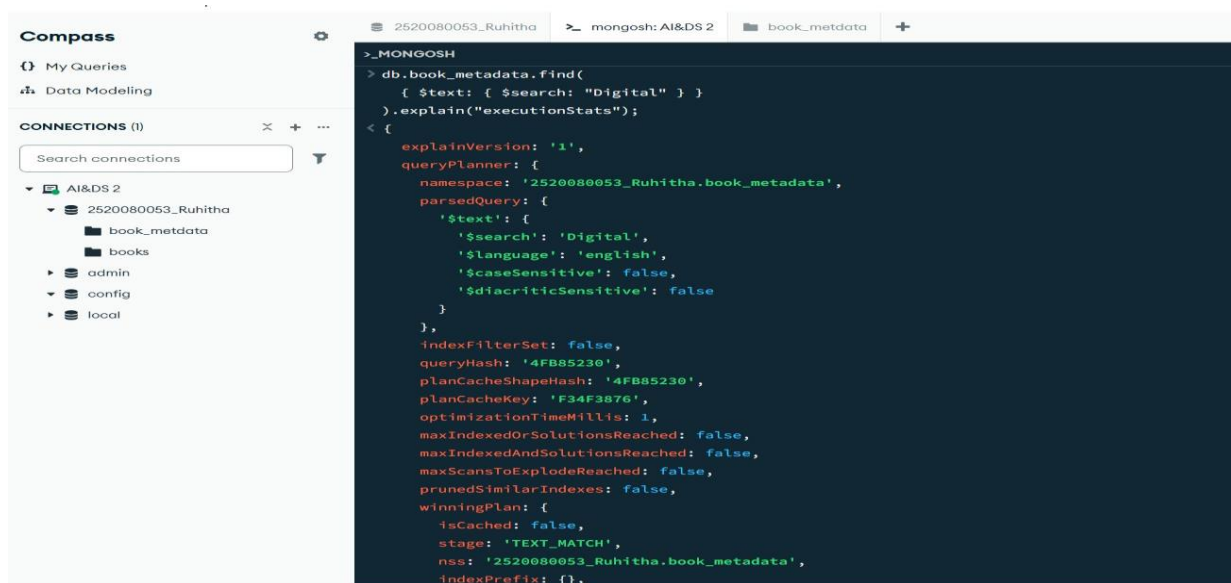
b) Query using the text index



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the database '2520080053_Ruhitha' and the collection 'book_metadata'. The main panel displays a MongoDB query in the 'MONGOSH' terminal:

```
>_MONGOSH  
{  
  totalReviews: 2,  
  author: 'R. Menon',  
  title: 'The Silent Algorithm',  
  avgReviewRating: 4.5  
}  
> db.book_metadata.createIndex({ "reviews.comment": "text" });  
< reviews.comment_text  
> db.book_metadata.find(  
  { $text: { $search: "Digital" } },  
  { title: 1, score: { $meta: "textScore" } }  
)<sort({ score: { $meta: "textScore" } }));  
< {  
  _id: ObjectId('6a8d20a1e9b1256610eda46f'),  
  title: 'The Silent Algorithm',  
  score: 1.1833333333333333  
}  
{  
  _id: ObjectId('6a8d217ee9b1256610eda472'),  
  title: 'Quantum Horizons',  
  score: 0.6  
}  
{  
  _id: ObjectId('6a8d217ee9b1256610eda470'),  
  title: 'Ocean of Code',  
  score: 0.6  
}  
}
```

c) Verify index usage with .explain()



The screenshot shows the MongoDB Compass interface. The main panel displays the output of the `.explain()` command in the 'MONGOSH' terminal:

```
>_MONGOSH  
> db.book_metadata.find(  
  { $text: { $search: "Digital" } }  
)<.explain("executionStats");  
< {  
  explainVersion: '1',  
  queryPlanner: {  
    namespace: '2520080053_Ruhitha.book_metadata',  
    parsedQuery: {  
      '$text': {  
        '$search': 'Digital',  
        '$language': 'english',  
        '$caseSensitive': false,  
        '$diacriticSensitive': false  
      }  
    },  
    indexFilterSet: false,  
    queryHash: '4FB85230',  
    planCacheShapeHash: '4FB85230',  
    planCacheKey: 'F34F3876',  
    optimizationTimeMillis: 1,  
    maxIndexedOrSolutionsReached: false,  
    maxIndexedAndSolutionsReached: false,  
    maxScansToExplodeReached: false,  
    prunedSimilarIndexes: false,  
    winningPlan: {  
      isCached: false,  
      stage: 'TEXT_MATCH',  
      nss: '2520080053_Ruhitha.book_metadata',  
      indexPrefix: {},  
    },  
  },  
}
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

2520080053_Ruhitha

book_metdata

books

admin

config

local

2520080053_Ruhitha

mongosh: AI&DS 2

book_metdata

+

>_MONGOSH

```
    indexName: 'reviews.comment_text',
    parsedTextQuery: {
      terms: [
        'digit'
      ],
      negatedTerms: [],
      phrases: [],
      negatedPhrases: []
    },
    textIndexVersion: 3,
    inputStage: {
      stage: 'FETCH',
      nss: '2520080053_Ruhitha.book_metdata',
      inputStage: {
        stage: 'IXSCAN',
        nss: '2520080053_Ruhitha.book_metdata',
        keyPattern: {
          _fts: 'text',
          _ftsx: 1
        },
        indexName: 'reviews.comment_text',
        isMultiKey: true,
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'backward',
        indexBounds: {}
      }
    }
  }
}
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

2520080053_Ruhitha

book_metdata

books

admin

config

local

2520080053_Ruhitha

mongosh: AI&DS 2

book_metdata

+

>_MONGOSH

```
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 6,
    totalKeysExamined: 3,
    totalDocsExamined: 3,
    executionStages: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      nss: '2520080053_Ruhitha.book_metdata',
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [
          'digit'
        ],
        negatedTerms: [],
        phrases: [],

```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

- 2520080053_Ruhitha
 - book_metdata
 - books
- admin
- config
- local

>_MONGOSH

```
negatedPhrases: []
},
textIndexVersion: 3,
docsRejected: 0,
inputStage: {
  stage: 'FETCH',
  nReturned: 3,
  executionTimeMillisEstimate: 0,
  works: 4,
  advanced: 3,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  nss: '2520080053_Ruhitha.book_metdata',
  docsExamined: 3,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
```

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

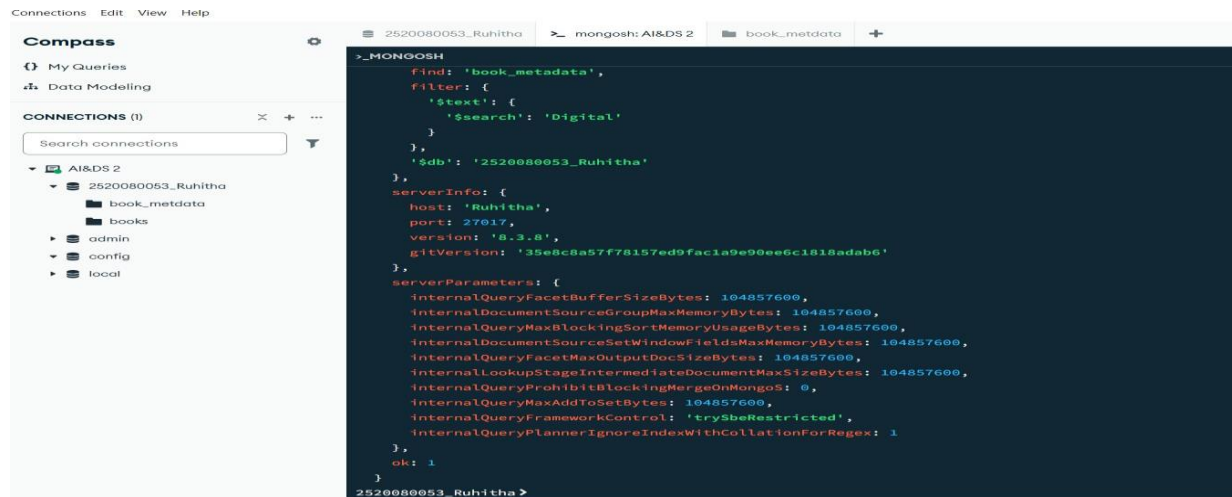
Search connections

AI&DS 2

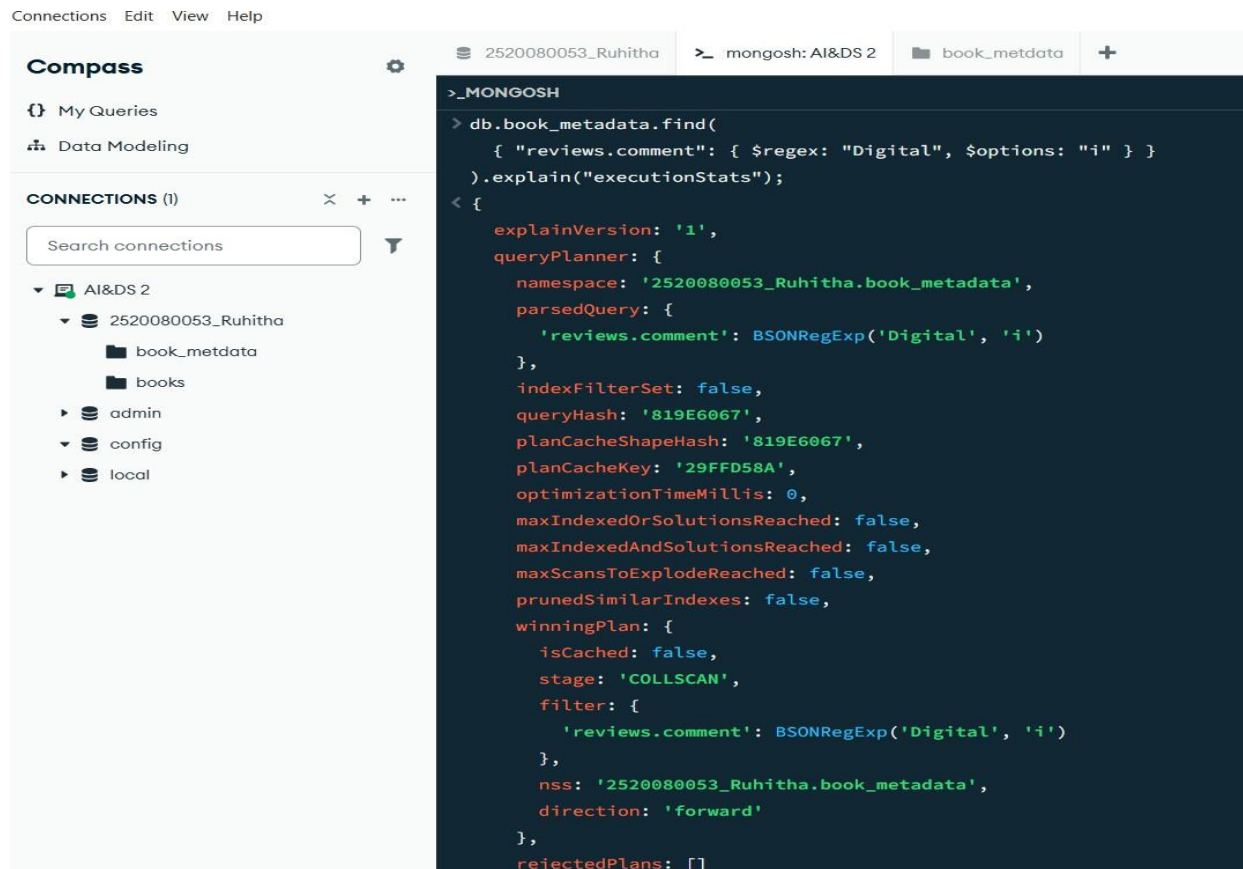
- 2520080053_Ruhitha
 - book_metdata
 - books
- admin
- config
- local

>_MONGOSH

```
nss: '2520080053_Ruhitha.book_metdata',
keyPattern: {
  _fts: 'text',
  _ftsx: 1
},
indexName: 'reviews.comment_text',
isMultiKey: true,
isUnique: false,
isSparse: false,
isPartial: false,
indexVersion: 2,
direction: 'backward',
indexBounds: {},
keysExamined: 3,
seeks: 1,
dupsTested: 3,
dupsDropped: 0,
peakTrackedMemBytes: 26
}
}
},
queryShapeHash: 'F3AD0A3A4DE07E9DFDF04007ABEE44B1166B74E6B4280F46E28830220186D407',
peakTrackedMemBytes: Long('26'),
command: {
  find: 'book_metdata',
  filter: {
    '$text': {
      '$search': 'Digital'
```



d) Compare against a non-indexed regex scan



Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

- 2520080053_Ruhitha
 - book_metadata
 - books
- admin
- config
- local

>_MONGOSH

```
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 1,
  totalKeysExamined: 0,
  totalDocsExamined: 4,
  executionStages: {
    isCached: false,
    stage: 'COLLSCAN',
    filter: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 5,
    advanced: 3,
    needTime: 1,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: '2520080053_Ruhitha.book_metadata',
    direction: 'forward',
    docsExamined: 4
  }
},
queryShapeHash: 'DE58F0EA1F9D0F0BEC392841E964F3E0A1D246F6636711090F560508D0DE2625',
command: {
  find: 'book_metadata',
```

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

AI&DS 2

- 2520080053_Ruhitha
 - book_metadata
 - books
- admin
- config
- local

>_MONGOSH

```
queryShapeHash: 'DE58F0EA1F9D0F0BEC392841E964F3E0A1D246F6636711090F560508D0DE2625',
command: {
  find: 'book_metadata',
  filter: {
    'reviews.comment': BSONRegExp('Digital', 'i')
  },
  '$db': '2520080053_Ruhitha'
},
serverInfo: {
  host: 'Ruhitha',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
2520080053_Ruhitha >
```

e) Supporting index for common queries (multikey index)

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the connection '2520080053_Ruhitha' under the 'AI&DS 2' cluster. The main panel displays the MongoDB shell output for the following commands:

```
>_MONGOSH
> db.book_metadata.createIndex({ formats: 1 });
db.book_metadata.createIndex({ avgRating: -1 });
db.book_metadata.find({ avgRating: { $gt: 4 } }).explain("executionStats");
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: '2520080053_Ruhitha.book_metadata',
    parsedQuery: {
      avgRating: {
        '$gt': 4
      }
    },
    indexFilterSet: false,
    queryHash: 'E21FCB34',
    planCacheShapeHash: 'E21FCB34',
    planCacheKey: '6A4442CC',
    optimizationTimeMillis: 1,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: '2520080053_Ruhitha.book_metadata',
      inputStage: {
        stage: 'IXSCAN',
        nss: '2520080053_Ruhitha.book_metadata',
        keyPattern: {
```

The screenshot shows the MongoDB Compass interface with the same connection. The main panel displays the execution statistics for the query:

```
>_MONGOSH
  avgRating: -1
},
  indexName: 'avgRating_-1',
  isMultiKey: false,
  multiKeyPaths: {
    avgRating: []
  },
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'forward',
  indexBounds: {
    avgRating: [
      '[inf, 4)'
    ]
  }
},
  rejectedPlans: []
},
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 45,
    totalKeysExamined: 3,
    totalDocsExamined: 3,
    executionStages: {
      isCached: false,
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- AI&DS 2
 - 2520080053_Ruhitha
 - book_metadata
 - books
 - admin
 - config
 - local

```

>_MONGOSH
  stage: 'FETCH',
  nReturned: 3,
  executionTimeMillisEstimate: 43,
  works: 4,
  advanced: 3,
  needTime: 0,
  needYield: 0,
  saveState: 1,
  restoreState: 1,
  isEOF: 1,
  nss: '2520080053_Ruhitha.book_metadata',
  docsExamined: 3,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 3,
    executionTimeMillisEstimate: 43,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 1,
    restoreState: 1,
    isEOF: 1,
    nss: '2520080053_Ruhitha.book_metadata',
    keyPattern: {
      avgRating: -1
    },
    indexName: 'avgRating_-1',
  }

```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

- AI&DS 2
 - 2520080053_Ruhitha
 - book_metadata
 - books
 - admin
 - config
 - local

```

>_MONGOSH
  find: 'book_metadata',
  filter: {
    avgRating: {
      '$gt': 4
    }
  },
  '$db': '2520080053_Ruhitha'
},
serverInfo: {
  host: 'Ruhitha',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
2520080053_Ruhitha>

```

